

PROBLEM STATEMENT

Square of Trigonometric Binomial

Find the derivative with respect to x of the function:

$$y = (1 + \cos 2x)^2$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক বর্গ ফাংশনটি উল্লেখ করি (Given Trigonometric Square Function)

$$y = (1 + \cos 2x)^2$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(1 + \cos 2x)^2]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^2] = 2u\frac{du}{dx}$)

$$\frac{dy}{dx} = 2(1 + \cos 2x)^{2-1} \cdot \frac{d}{dx}(1 + \cos 2x) = 2(1 + \cos 2x) \cdot \frac{d}{dx}(1 + \cos 2x)$$

STEP 4 বন্ধনীভুক্ত অংশের অন্তরক নির্ণয় করি (Differentiating Inner Sum)

$$\frac{d}{dx}(1 + \cos 2x) = 0 - \sin 2x \cdot \frac{d}{dx}(2x) = -2 \sin 2x$$

STEP 5 মান বসিয়ে ধ্রুবক গুণ করে চূড়ান্ত অন্তরজ পাই (Multiplying Constants)

$$\frac{dy}{dx} = 2(1 + \cos 2x) \cdot (-2 \sin 2x) = -4 \sin 2x(1 + \cos 2x)$$

FINAL ANSWER
$$\frac{dy}{dx} = -4 \sin 2x(1 + \cos 2x)$$

📌 BOX 1: FORMULATION & DOUBLE-ANGLE EQUIVALENCES

- **Standard Chain Result:** $\frac{dy}{dx} = -4 \sin 2x(1 + \cos 2x)$.
- **Half-Angle Identity Transformation:** Since $1 + \cos 2x = 2 \cos^2 x$:

$$y = (2 \cos^2 x)^2 = 4 \cos^4 x$$

$$\frac{dy}{dx} = 4 \cdot 4 \cos^3 x (-\sin x) = -16 \sin x \cos^3 x = -8 \sin 2x \cos^2 x$$

- Both representations are completely equivalent.

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Missing Factor of 2 in Argument:** Differentiating $\cos 2x$ to $-\sin 2x$ without multiplying by the inner derivative $\frac{d}{dx}(2x) = 2$.
- **Dropping the Negative Sign:** Remembering that $\frac{d}{dx}(\cos 2x) = -2 \sin 2x$.

🌐 BOX 3: REAL DOMAIN & PERIODICITY

- **Domain & Range:** Smooth and defined on $\mathbb{R} = (-\infty, \infty)$. Since $0 \leq 1 + \cos 2x \leq 2$, the function range is $[0, 4]$.
- **Periodicity:** The fundamental period of $y = (1 + \cos 2x)^2$ is $T = \pi$.
- **Stationary Points:** $\frac{dy}{dx} = 0$ when $\sin 2x = 0 \implies x = \frac{k\pi}{2}$.

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = 1 + \cos 2x \implies y = u^2$.
- $\frac{dy}{du} = 2u$.
- $\frac{du}{dx} = -2 \sin 2x$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 2u \cdot (-2 \sin 2x) = -4 \sin 2x(1 + \cos 2x)$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct power rule setup $2(1 + \cos 2x) \cdot \frac{d}{dx}(1 + \cos 2x)$.
- **1.0 Mark:** Accurate computation of inner derivative $-2 \sin 2x$.
- **1.0 Mark:** Clean combination and finalized boxed expression $-4 \sin 2x(1 + \cos 2x)$.